Fundamental Physics 1: Mechanics

Nguyen Thi Hong Van

INSTITUTE OF PHYSICS, HANOI

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Major sources

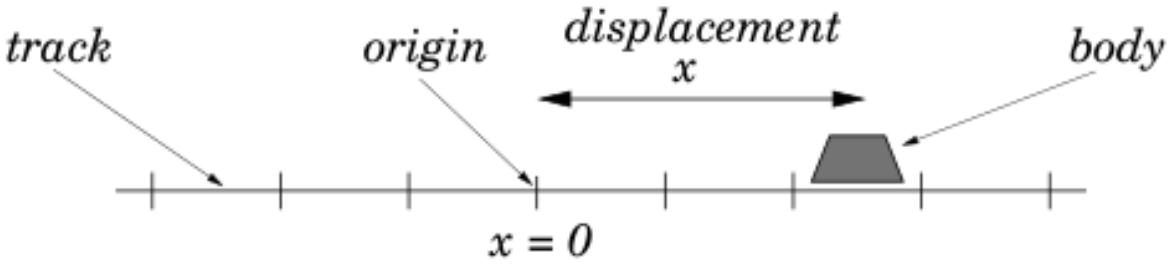
Major sources

1. Halliday and Resnick, Fundamentals of PHYSICS 10th Edition, Jearl Walker
2. G.R. Fowles, Third edition (Holt, Rinehart, & Winston, New York NY, 1977).

Chapter 1: Overview of Classical Mechanics (CM)

- CM is the study of the motion of bodies in accordance with the general principals first enunciated by Issac Newton (1687)
- CM has many applications in different areas of science such as **astronomy**, **Chemistry**, **Geology**, and **Engineering**
- Types of motion studied in CM:
  - **Translational motion**: a body shifts from one point in space to another
  - **Rotational motion**: an extended body changes orientation, with respect to other bodies in space, without changing position
  - **Oscillatory motion**: continually repeats in time with a fixed period
  - **Circular motion**: a body executes a circular orbit about another fixed body.

Displacement



- Distance travelled between t_1 and t_2 (if the direction of motion does not change)

$$S = |x(t_2) - x(t_1)|$$

Displacement and Distance



Combining displacements

Combining displacements

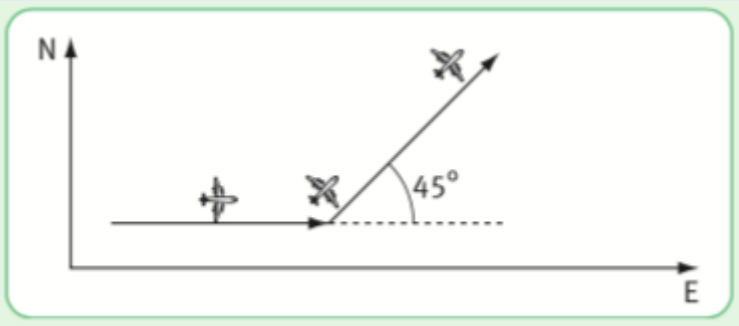


- Distance travelled = 25 km
- Displacement = 15 km north-east

Combining displacements

Worked example

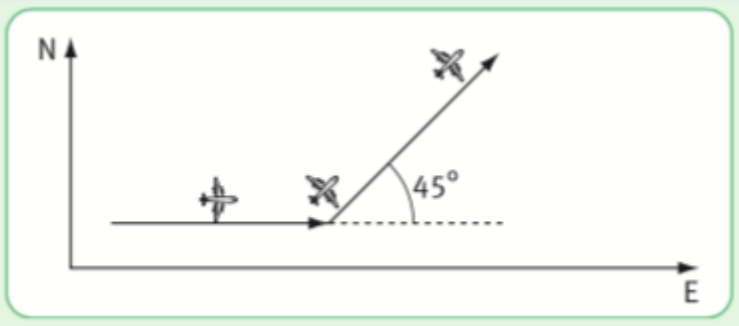
An aircraft flies 30 km due east and then 50 km north-east (see the Figure). Calculate the final displacement and the distance traveled of the aircraft.

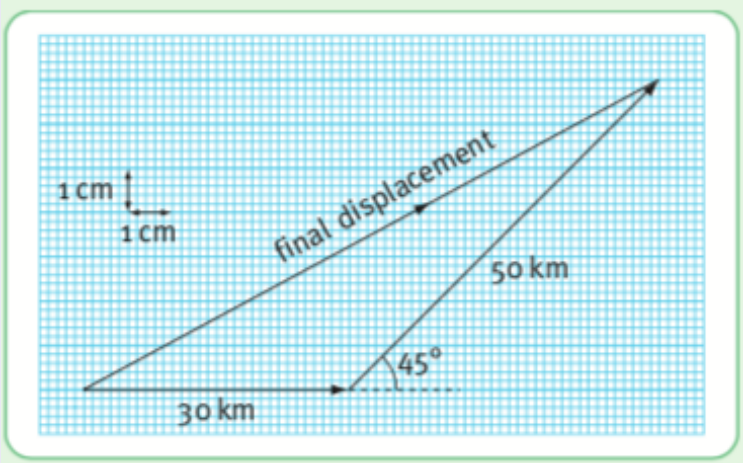


Combining displacements

Worked example

An aircraft flies 30 km due east and then 50 km north-east (see the Figure). Calculate the final displacement and the distance traveled of the aircraft.





Final displacement = 74 km
Angle = 28° N of E
Distance traveled: 80 km

Velocity

- Velocity is the rate of change of displacement with time
- Average velocity:

$$\bar{v} = \frac{x(t_2) - x(t_1)}{t_2 - t_1} = \frac{\Delta x}{\Delta t}$$

- Instantaneous velocity:

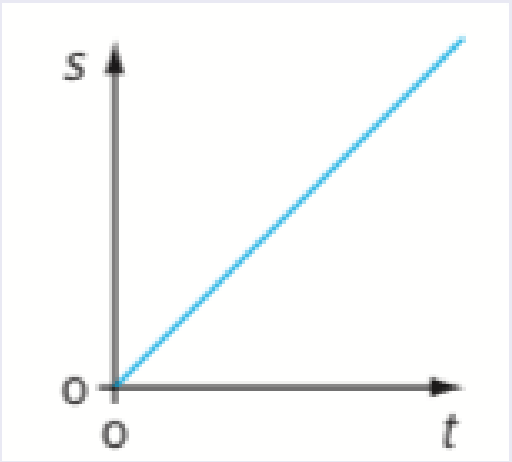
$$v = lim_{\Delta t \to 0} \frac{\Delta x}{\Delta t} = \frac{dx}{dt}$$

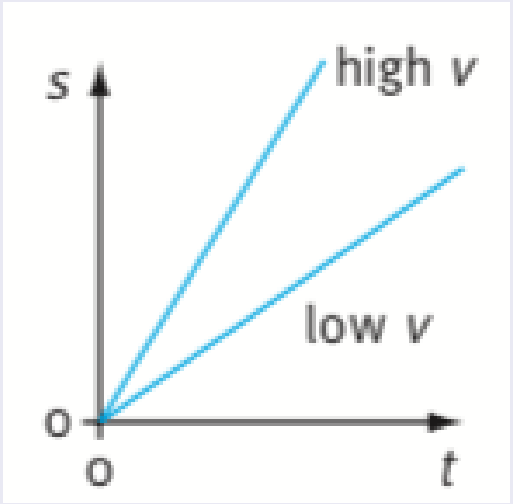
- Average speed: the ratio of the total distance it travels to the total time it takes to travel that distance.
- Instantaneous speed $= |v|$

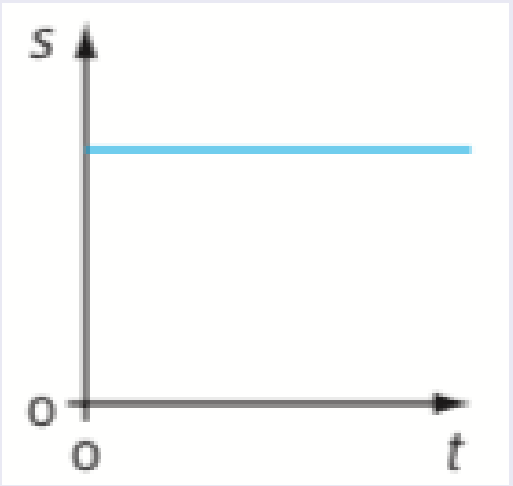
Worked example:

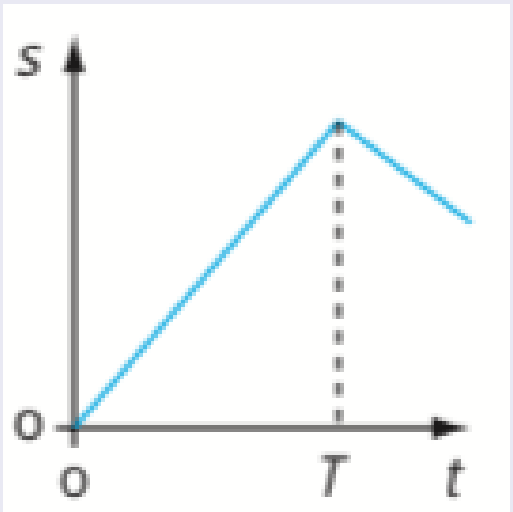
1. Given $v = const$, find the dependence of displacement x on time t?
2. Plot the graph of displacement versus time for a body moving with constant velocity?

Displacement - Time graph: Motion with constant velocity

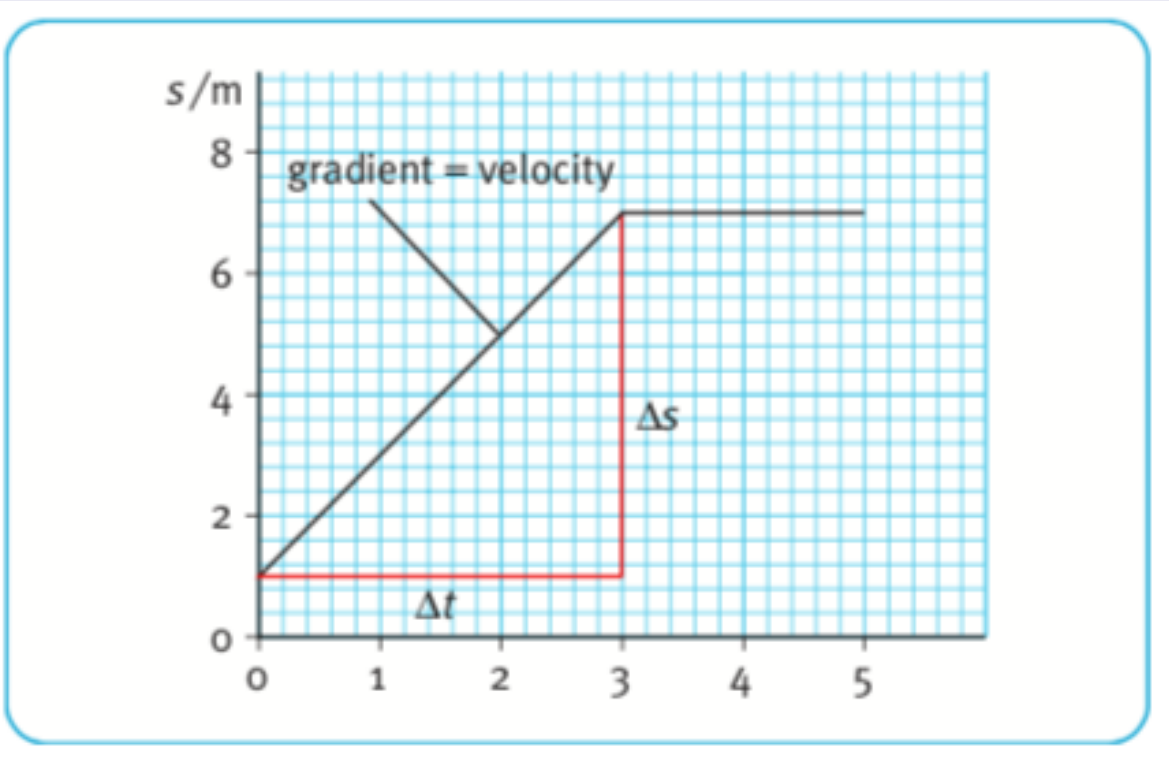




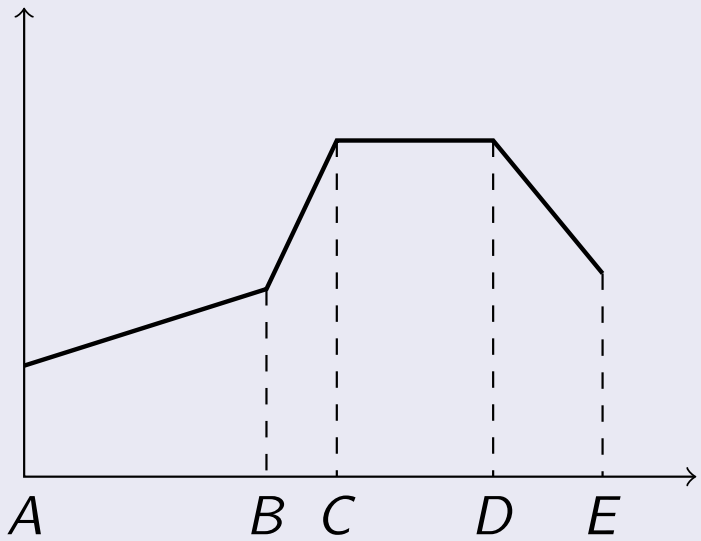




Deducing velocity from a displacement–time graph

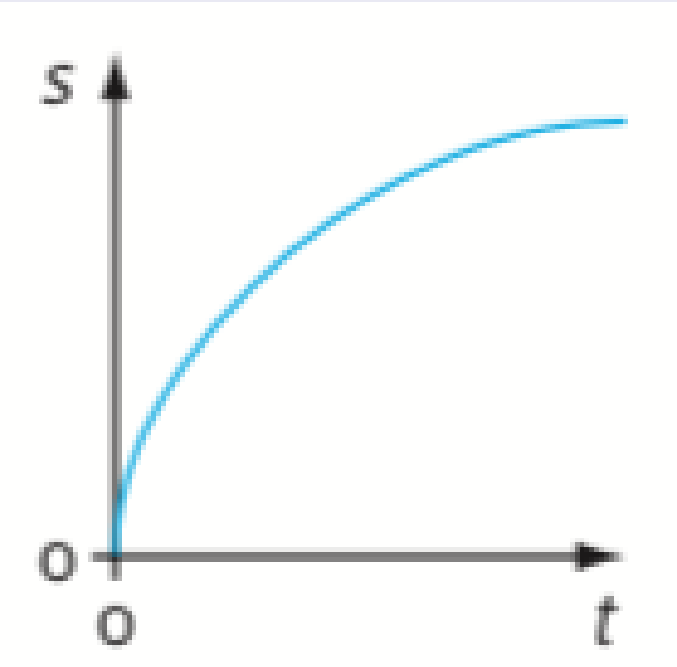


Motion with constant velocity



- The graph of x versus t is a straight line : $x(t) = x_0 + v.t$
- $v = \Delta x/\Delta t \rightarrow$ gradient of the graph/straight line.
- x increases $\rightarrow v > 0$ OR object moving to the right,
- x decreases $\rightarrow v < 0$ OR object moving to the left,
- $x = \text{const} \rightarrow v = 0$ OR object remains at rest.

Displacement - Time graph: Motion with velocity is changing



Acceleration

- Acceleration is the rate of the change of velocity with time
- Average value:

$$\bar{a} = \frac{\Delta v}{\Delta t}$$

- Instantaneous value:

$$a = lim_{\Delta t \to 0} \frac{\Delta v}{\Delta t} = \frac{dv}{dt} = \frac{d^2x}{dt^2}$$

Acceleration and deceleration

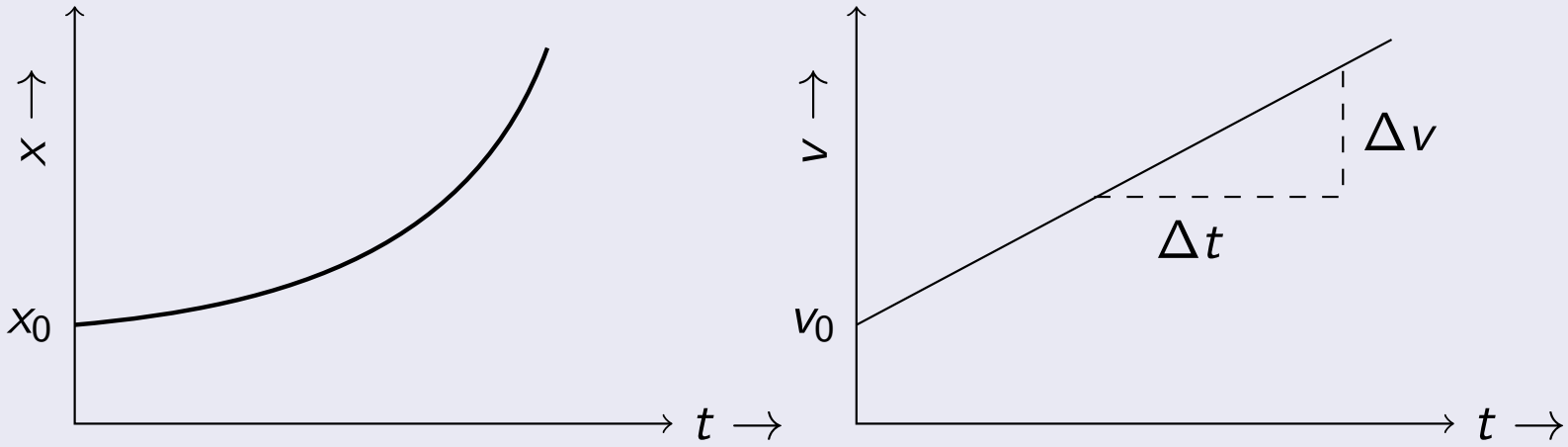
Acceleration and deceleration

- When the speed increases, we have the so-called "acceleration" and acceleration and velocity is in the same direction ($a.v > 0$).
- When the speed decreases, acceleration and velocity is in opposite direction ($a.v < 0$) and we have so-called "deceleration" or "retardation"

Equation of kinematics (for motion with a constant acceleration)

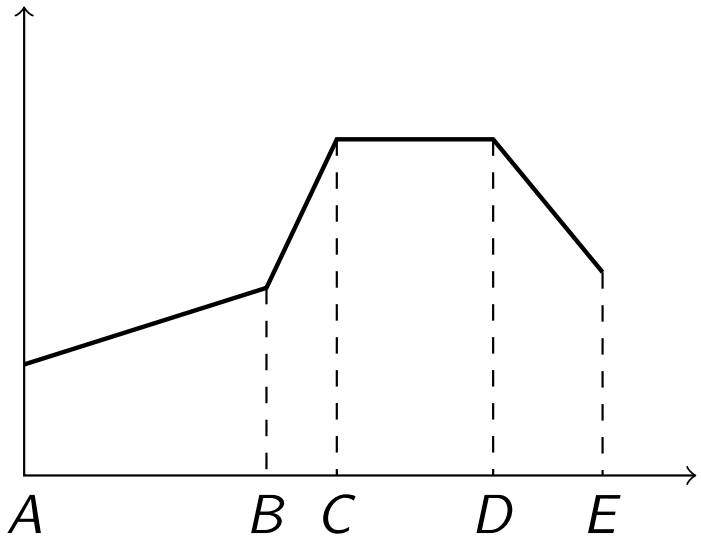
- $v(t) = v_0 + a.t$
- $x(t) = x_0 + v_0.t + a.t^2/2$
- $v(t)^2 - v_0^2 = 2.a.(x(t) - x_0)$

Motion with constant acceleration



- $a = dv/dt \rightarrow dv = a.dt \rightarrow v(t) = v_0 + a.t \rightarrow$ straight line
- $dx = v.dt \rightarrow x = x_0 + v_0.t + a.t^2/2 \rightarrow$ parabola.
- $S = x - x_0 = v_0.t + a.t^2/2 \rightarrow$ the net distance traveled after t seconds
- $v^2(t) = v_0^2 + 2.a.S$

Motion in one dimension: Graphical analysis of motion

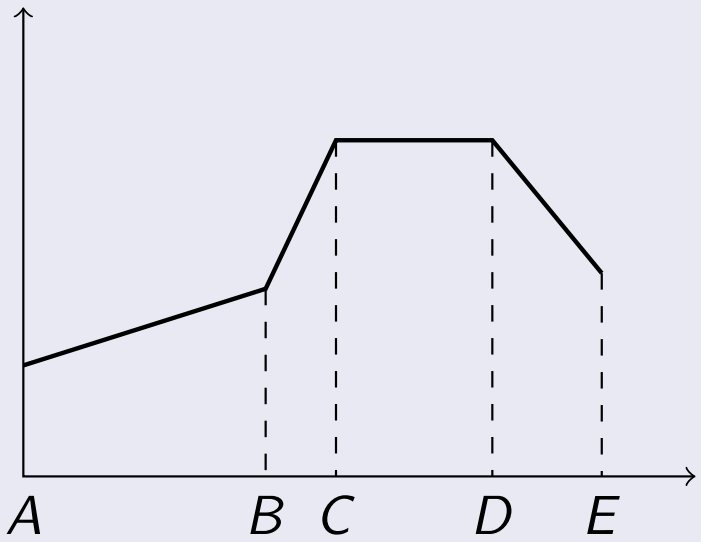


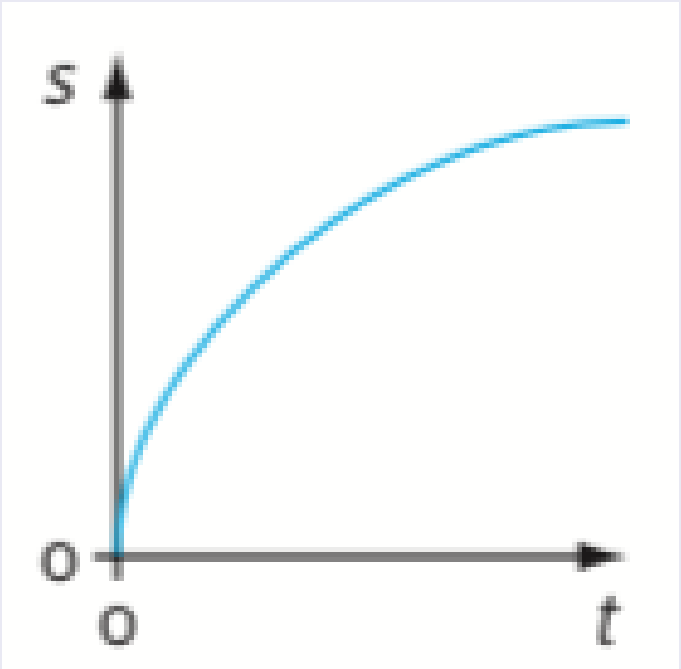
Displacement - time graph

- Gradient of graph is the velocity of the body.
- If the gradient of graph is positive $\Rightarrow$ direction of motion is in the positive direction of displacement x.
- If the gradient of graph is negative $\Rightarrow$ direction of motion is in opposite direction of the displacement x.

Displacement - time graph

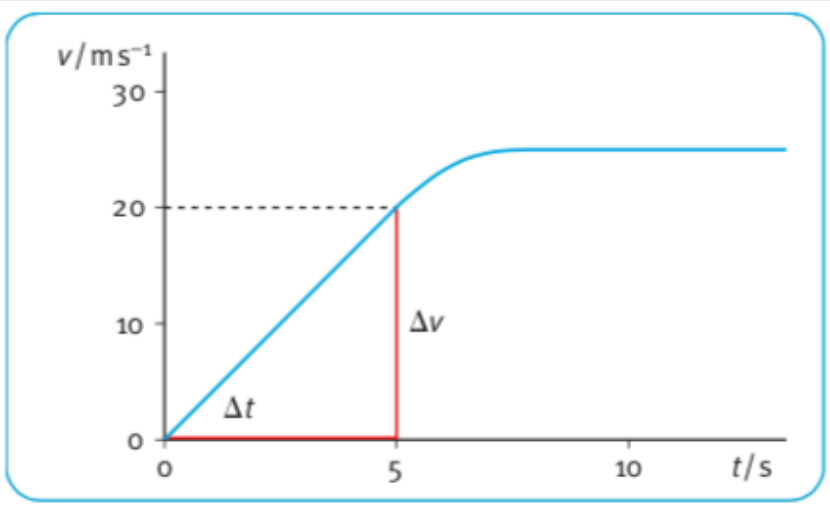
Displacement - time graph

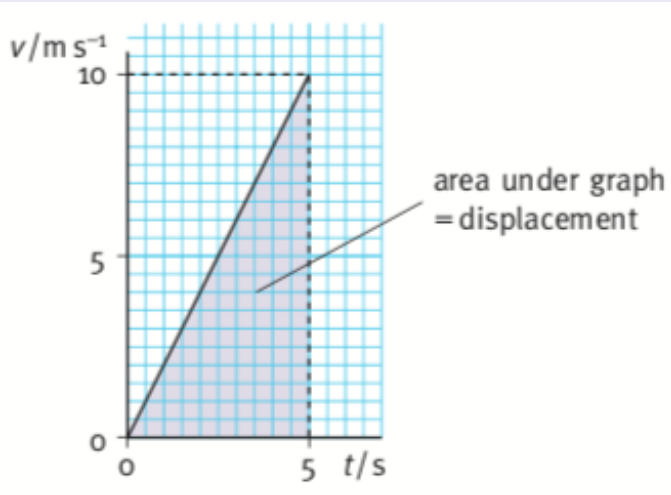




- Horizontal line – *body at rest*.
- Straight line with positive gradient – *uniform velocity*.
- Straight line with negative gradient – *uniform velocity in the opposite direction*.
- Curve – *non-uniform velocity*

Velocity-time graph





Velocity - time graph

- Gradient of graph is the acceleration of the body.
- Area under the graph gives the displacement of the body.

Velocity - time graph

Examples: sketch graphs of $(v - t)$ to describe the following motions:

1. Body at rest
2. Body moves with uniform velocity of 5 m/s.
3. Body moves from rest with uniform acceleration.
4. Initial velocity $\neq 0$.
5. Graph with negative gradient.
6. Non-uniform acceleration
   - Acceleration increases
   - Acceleration decreases
   - Deceleration increases
   - Deceleration decreases

Motion in one dimension: Free fall under gravity (1)

- The gravitational force of the Earth pulls all objects towards the centre of the Earth.
- A body is in free fall motion if it moves under the influence of the Earth's gravity alone.
- Without air resistance, a body in free fall has an acceleration known as the acceleration of free fall g.
- Galileo: all bodies in free-fall close to the Earth surface accelerate vertically downward with the same acceleration: $g = 9.81 m/s^2$

Motion in one dimension: Free fall under gravity (2)

???? Sketch **velocity - time graph** and **distance - time graph** for a body under free fall from rest.

Motion in one dimension: Free fall under gravity (3)

??? Sketch **velocity - time graph** and **distance - time graph** for a body projected vertically upwards.

Worked example

A stone is thrown vertically upwards from a point that is 12m above the sea. It then falls into the sea below after 3.4 s. Air resistance is negligible. At which speed was the stone released when it was thrown?

A $3.5\ \text{m s}^{-1}$

B $6.6\ \text{m s}^{-1}$

C $13\ \text{m s}^{-1}$

D $20\ \text{m s}^{-1}$

Worked example

A stone is thrown vertically upwards from a point that is 12m above the sea. It then falls into the sea below after 3.4 s. Air resistance is negligible. At which speed was the stone released when it was thrown?

A $3.5\ \text{m s}^{-1}$

B $6.6\ \text{m s}^{-1}$

C $13\ \text{m s}^{-1}$

D $20\ \text{m s}^{-1}$

Solution

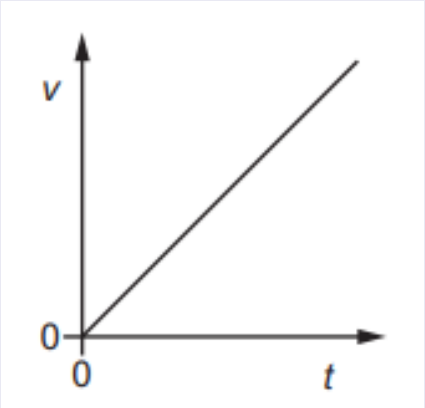
C is correct

Chapter 1: Effect of air resistance to free-fall motion

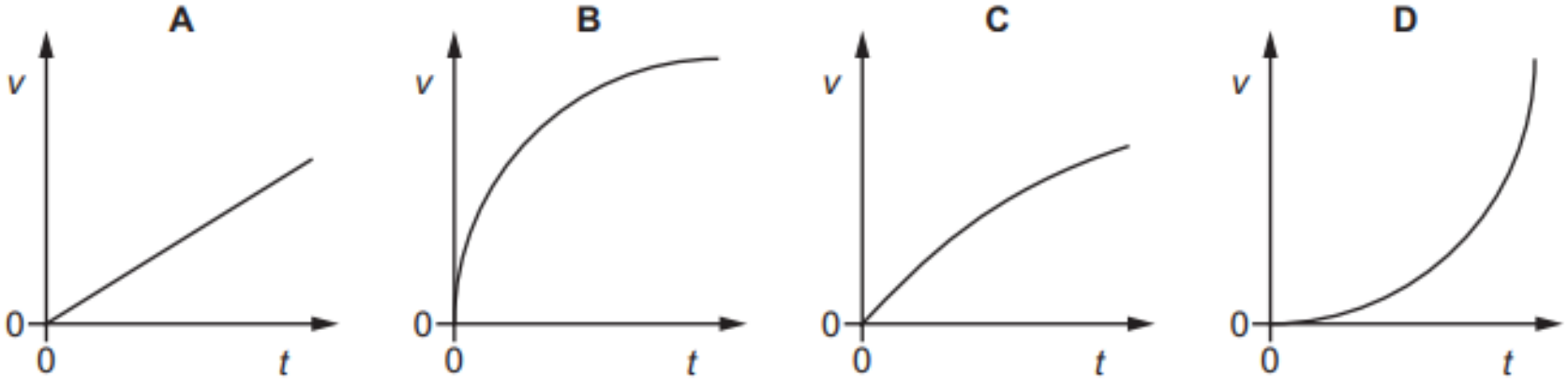
- When an object falls through the air, it experiences air resistance, which opposes its motion.
- Because if air resistance, the acceleration of an object falling in air is less than the acceleration of free fall.
- The air resistance increases when the velocity of the object increases.
- For an object which is released from rest and falls in air:
  - Initially its velocity increases,
  - As its velocity increases, the air resistance increases,
  - The resultant acceleration of the object decreases.
  - Finally the acceleration becomes zero and it falls at a constant velocity known as the **terminal velocity**
- A heavier sphere would have a larger terminal velocity than a lighter sphere.

Worked example

An object falls freely from rest in a vacuum. The graph shows the variation with time t of the velocity v of the object.



Which graph, **using the same scales**, represents the object falling in air?



Solution

C is correct

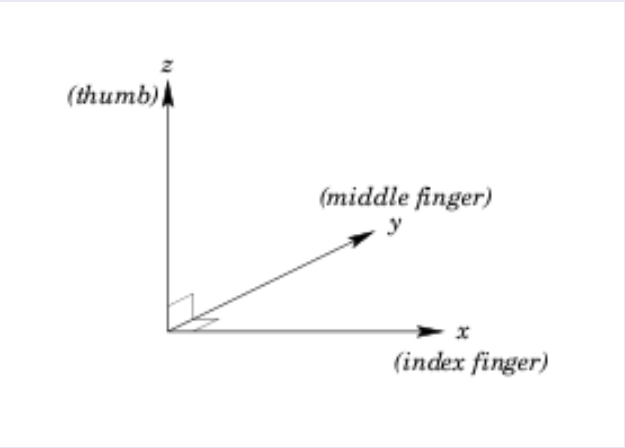
Chapter 1: Motion in three dimensions (1)

To deal with 3-dimensional motion, it is necessary to set up a suitable coordinate system.

Vectors

- Cartesian coordinates

Cartesian coordinate system consists of three mutually perpendicular axes.



- Vector and scalar quantities
  - Scalar quantity is specified by a single value with an appropriate unit and has no direction such as temperature, volume, mass, time
  - A vector quantity has both magnitude and direction. Examples: displacement,velocity, accelerator, force.

Some properties of vector

- Equality of two vectors:
  $\vec{A} = \vec{B} \Leftrightarrow |\vec{A}| = |\vec{B}|$ & $\vec{A}$ and $\vec{B}$ point in the same direction.
- Vector addition
  - Sum of two vectors is the vector down from the tail of the first vector to the tip of the last vector.
  - Parallelogram rule of addition $\rightarrow \vec{A} + \vec{B} = \vec{B} + \vec{A}$
  - Associative law of addition: $\vec{A} + (\vec{B} + \vec{C}) = (\vec{A} + \vec{B}) + \vec{C}$
- Negative of a vector: $\vec{A} + (-\vec{A}) = 0 \rightarrow$ have the same magnitude but point in opposite direction.
- Subtracting vector: $\vec{A} - \vec{B} = \vec{A} + (-\vec{B})$

Components of a vector and unit vectors

- Components of a vector are the projections of that vector along coordinate axes: $\vec{A} = (A_x, A_y, A_z)$.
- ?? Can the components of a vector ever be greater than the magnitude of the vector?
- hint: $A_x = |\vec{A}|.sin\theta.cos\phi$; $A_y = |\vec{A}|.sin\theta \sin\phi$; $A_z = |\vec{A}|.cos\theta$

Components of a vector and unit vectors

- Unit vectors
  - a unit vector is a dimensionless vector having a magnitude of exactly 1. It is used to specify a given direction and have no other physical significance:
    - $\vec{i} \rightarrow x$; $\vec{j} \rightarrow y$; $\vec{k} \rightarrow z$.
    - $|\vec{i}| = |\vec{j}| = |\vec{k}| = 1$
    - $\vec{i}.\vec{j} = \vec{i}.\vec{k} = \vec{j}.\vec{k} = 0$
    - $\vec{A} = A_x.\vec{i} + A_y.\vec{j} + A_z.\vec{k}$.
  - Vector magnitude: $A = \sqrt{A_x^2 + A_y^2 + A_z^2}$
- Scalar multiplication:
  - $\vec{S} = \lambda.\vec{r}$, λ is a number/scalar
  - $\Rightarrow \vec{S} = \lambda(x, y, z) = (\lambda x, \lambda y, \lambda z)$: multiplying all the components of the original.
  - Scalar product of two vectors:

$$\vec{A}.\vec{B} = A_x.B_x + A_y.B_y + A_z.B_z$$

Vector displacement, vector velocity and vector acceleration

- Vector displacement: $\vec{r}(t) = [x(t), y(t), z(t)]$
- Vector velocity:

$$\vec{v}(t) = lim_{\Delta t \to 0} \frac{\vec{r}(t + \Delta t) - \vec{r}(t)}{\Delta t} = \frac{d\vec{r}}{dt}$$

$$v_x = \frac{dx}{dt}; \quad v_y = \frac{dy}{dt}; \quad v_z = \frac{dz}{dt}$$

- Vector acceleration:

$$\vec{a}(t) = lim_{\Delta t \to 0} \frac{\vec{v}(t + \Delta t) - \vec{v}(t)}{\Delta t} = \frac{d\vec{v}}{dt} = \frac{d^2\vec{r}}{dt^2}$$

$$a_x = \frac{dv_x}{dt} = \frac{d^2x}{dt^2}$$

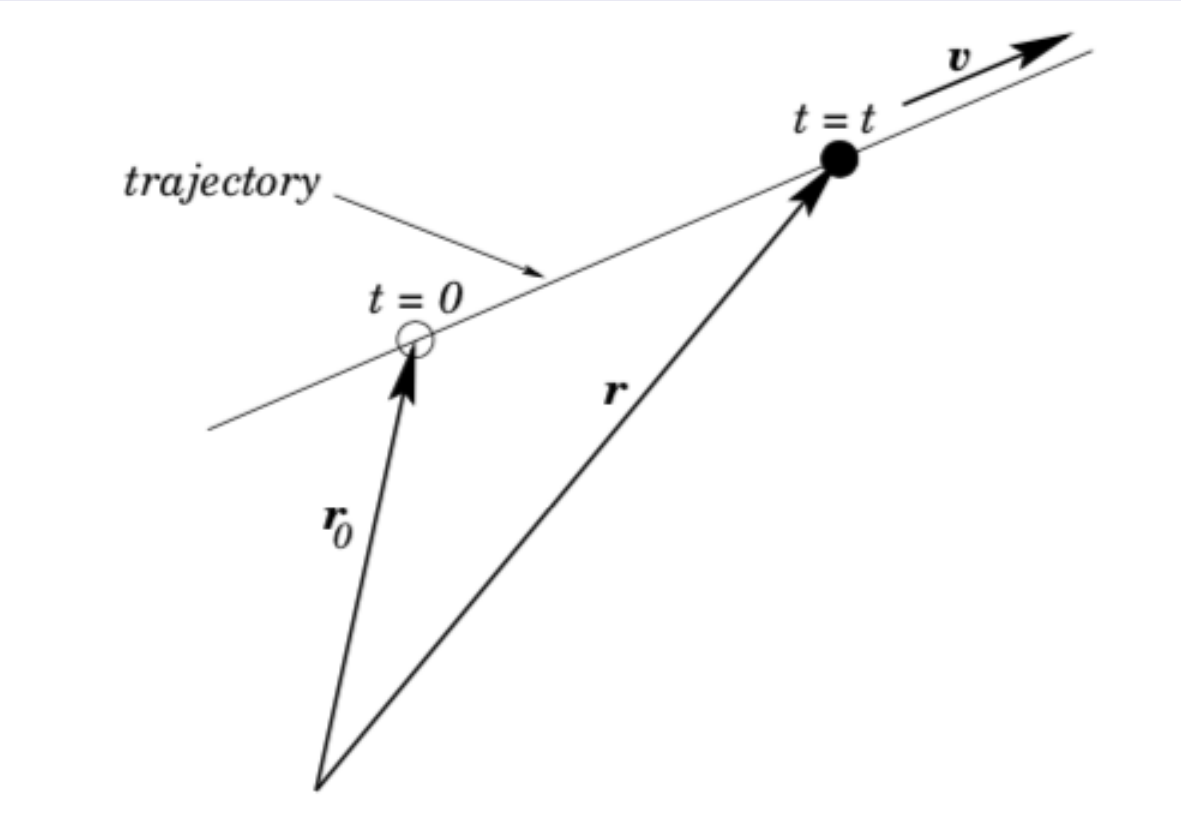
$$a_y = \frac{dv_y}{dt} = \frac{d^2y}{dt^2}$$

$$a_z = \frac{dv_z}{dt} = \frac{d^2z}{dt^2}$$

Motion with constant velocity

- $\vec{r}(t) = \vec{r}_0 + \vec{v}.t$
- The trajectory of the object is parallel to $\vec{v} \Rightarrow$ a straight-line
- In components:

$$x(t) = x_0 + v_x.t$$



Motion with constant acceleration

- $\vec{r}(t) = \vec{r_0} + \vec{v_0}.t + \vec{a}.t^2/2$
- $\vec{S} = \vec{v_0}.t + \vec{a}.t^2/2$
- $\vec{v}(t) = \vec{v_0} + \vec{a}.t$
- $v(t)^2 = v_0^2 + 2\vec{a}.\vec{S}$

In components:

$$x(t) = x_0 + v_{0x}.t + \frac{a_x.t^2}{2}$$

$$v_x(t) = v_{0x} + a_x.t$$

Motion with constant acceleration

- $\vec{r}(t) = \vec{r_0} + \vec{v_0}.t + \vec{a}.t^2/2$
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In components:

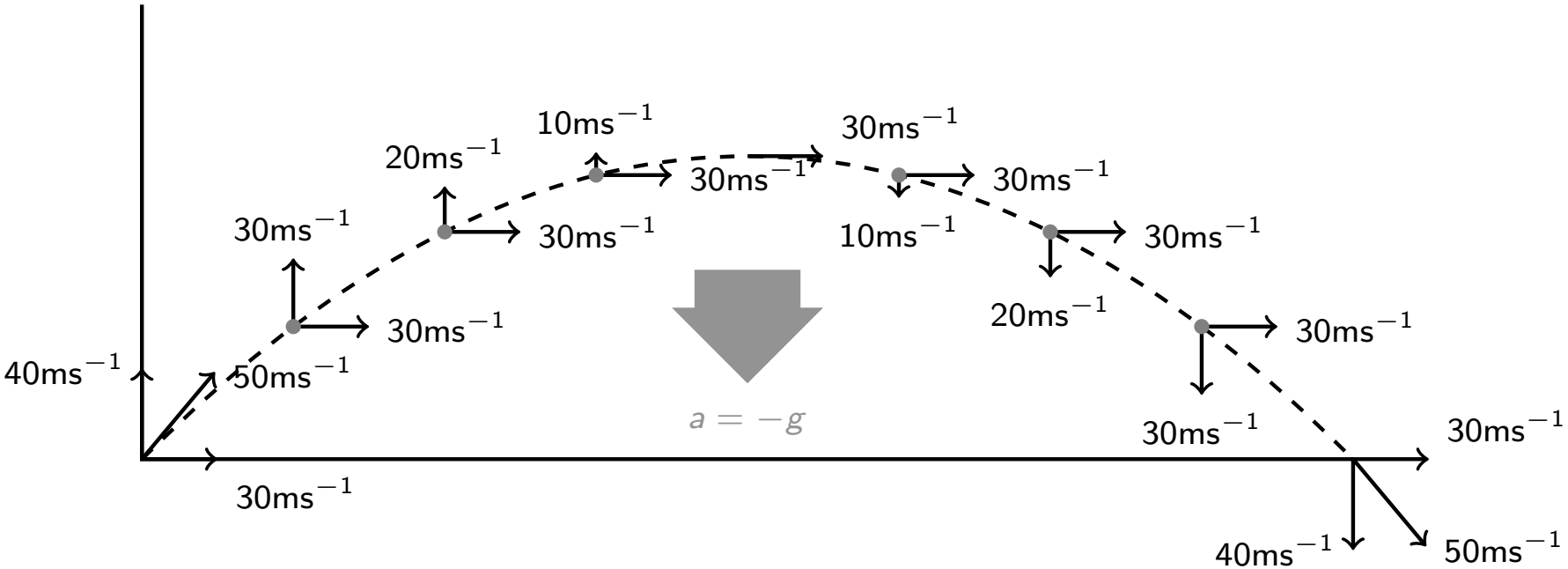
$$x(t) = x_0 + v_{0x}.t + \frac{a_x.t^2}{2}$$

$$v_x(t) = v_{0x} + a_x.t$$

Examples

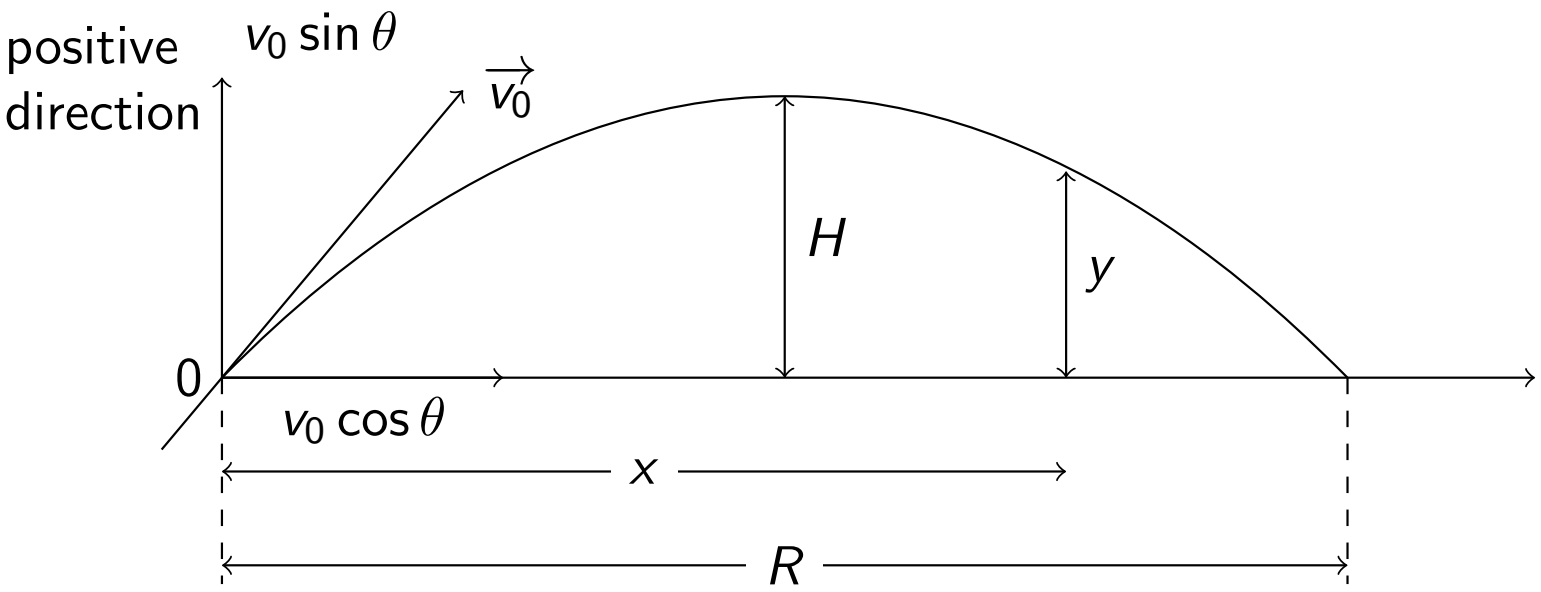
- Free-fall motion under gravity
- Projectile motion

Chapter 1: Projectile motion (1)



- The horizontal component of the velocity is constant
- The vertical component of the velocity changes
- The vertical component of the velocity is zero at the highest point
- The vertical acceleration $a = -g$ throughout the motion

Chapter 1: Projectile motion (2)



Projectile motion

- Given: $\vec{a} = (0, -g, 0)$; $\vec{v_0} = (v_0.\cos\theta, v_0.\sin\theta), 0$
- ?? Find horizontal range $R = v_0^2.sin2\theta/g \Rightarrow R_{max} = v_0^2/g$ when $\theta = 45^\circ$
- ?? Find altitude of the projectile $H = v_0^2.sin^2\theta/2g \Rightarrow h_{max} = v_0^2/2g$ when $\theta = 90^\circ$

Worked example

A stone is thrown with a velocity of 8 m/s into the air at an angle of 40° to the horizontal.

(a) Calculate the vertical component of the velocity.

(b) State the value of the vertical component of the velocity when the stone reaches its highest point. Ignore air resistance.

(c) Use your answers to (a) and (b) to calculate the time the stone takes to reach it highest point.

(d) Calculate the horizontal component of the velocity.

(e) Use your answers to (c) and (d) to find the horizontal distance travelled by the stone while climbing to its highest point.

Result

(a) 5.1 m/s

(b) 0

(c) 0.52 s

(d) 6.1 m/s

(e) 3.2 m

Worked example

A stone is thrown horizontally from the top of a vertical cliff and lands 4.0 s later at a distance 12.0 m from the base of the cliff. Ignore air resistance.

(a) Calculate the horizontal speed of the stone.

(b) Calculate the height of the cliff.

Worked example

A stone is thrown horizontally from the top of a vertical cliff and lands 4.0 s later at a distance 12.0 m from the base of the cliff. Ignore air resistance.

(a) Calculate the horizontal speed of the stone.

(b) Calculate the height of the cliff.

Result

(a) 3.0 m/s

(b) 78.5 m

Projectile Motion (6)

A skydiver falls from an aircraft that is moving horizontally. The vertical component of the velocity of the skydiver is v. The vertical component of the acceleration of the skydiver is a.
Which row describes v and a during the first few seconds after the skydiver leaves the aircraft?

	v	a
A	constant	constant
B	constant	decreasing
C	increasing	constant
C	increasing	decreasing

Projectile Motion (6)

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Which row describes v and a during the first few seconds after the skydiver leaves the aircraft?

	v	a
A	constant	constant
B	constant	decreasing
C	increasing	constant
C	increasing	decreasing

Solution

D is correct

Relative motion

- $\vec{v}_{12} \rightarrow$ velocity of object (1) with respect to frame (2)
- $\vec{v}_{23} \rightarrow$ velocity of frame (2) with respect to frame (3)
- $\vec{v}_{13} \rightarrow$ velocity of object (1) with respect to frame (3)
- Relative velocity: $\vec{v}_{13} = \vec{v}_{12} + \vec{v}_{23}$
- Relative position/displacement: $\vec{r}_{13} = \vec{r}_{12} + \vec{r}_{23}$
- Relative acceleration: $\vec{a}_{13} = \vec{a}_{12} + \vec{a}_{23}$

Relative motion

Worked example

The speed of an aircraft in still air is 200 km h^{-1}. The wind blows from the west at a speed of 85.0 km h^{-1}.
In which direction must the pilot steer the aircraft in order to fly due north?

A 23.0° east of north

B 23.0° west of north

C 25.2° east of north

D 25.2° west of north

Worked example

The speed of an aircraft in still air is 200 km h^{-1}. The wind blows from the west at a speed of 85.0 km h^{-1}.
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B 23.0° west of north

C 25.2° east of north

D 25.2° west of north

Solution

D is correct